

Retrieval-Augmented Generation (RAG) and Large Language Models

What are Large Language Models?

Large Language Models (LLMs) are neural networks trained on massive text corpora to predict the next token in a sequence. Models like GPT-4, Claude, Gemini, and Llama have billions of parameters and can generate coherent, contextually relevant text.

LLMs exhibit emergent capabilities: reasoning, code generation, translation, summarisation, and question answering — all from a single pre-trained model fine-tuned on instructions.

Transformer Architecture

The Transformer (Vaswani et al., 2017) introduced self-attention, allowing every token to attend to every other token in the sequence. Positional encodings preserve word order.

Key components: multi-head self-attention, feed-forward layers, layer normalisation, residual connections, and (in decoders) cross-attention over encoder outputs.

What is RAG?

Retrieval-Augmented Generation (RAG) combines a retrieval system with a generative LLM. Instead of relying solely on the model's parametric knowledge, RAG fetches relevant documents from an external knowledge base and includes them in the prompt context.

RAG addresses LLM hallucination, knowledge cutoff, and the inability to cite sources — making it ideal for enterprise knowledge bases, customer support, and research assistants.

RAG Pipeline

Indexing phase: Documents are split into chunks, embedded with a sentence encoder, and stored in a vector store (FAISS, Pinecone, Weaviate, Chroma).

Query phase: The user query is embedded, a similarity search retrieves the top-k chunks, and they are injected into the LLM prompt as context before generation.

Vector Databases

Vector databases store high-dimensional embeddings and support approximate nearest neighbour (ANN) search. FAISS (Meta) is an open-source library for efficient similarity search; Pinecone and Weaviate are managed cloud services.

Common distance metrics: cosine similarity (angle between vectors), L2 (Euclidean), and dot product. For normalised embeddings cosine and dot product are equivalent.

Best Practices

Chunk size matters: too small loses context; too large dilutes relevance. 256-512 tokens with 10-15% overlap is a good starting point.

Re-ranking: after retrieval, a cross-encoder re-ranks candidates for higher precision.

Hybrid search: combine dense (vector) retrieval with sparse (BM25/keyword) retrieval.

Evaluate with RAGAS metrics: faithfulness, answer relevancy, context precision & recall.